a) Identify and explain the four major components of the IOT system mangement Architecture in the smart Agriculture scenario.

Ans:- 1] Manged Devices: The 50 temperature sensors that collect temperature readings.

2] Managment Agent: Run on the IOT gateway and mangens devices configurations.

3] Network Management system (NMS): Allows the administator to configure and monitor all sensors centrally.

4] Management protocol: NETCONF manages device configrations, while YANG defins the configuration data structure.

b) Explain How NETCONF and YANG work togather to automate the configuration of the temperature sensors. Identify the appropriate NETCONF operation for changing the reporting interval to 30 seconds.

Ans: NETCONF is a Network configuration protocol that allows administrators to configure devices remotly while YANG defines the structure and rules of the configuration data.

NETCONF uses YANG data models to configure the temperature sensor consistently and automatically.

2

The administrator can set the reporting interval to 30 second and the temperature threshold to 35°C.

The appropriate NETCONF operating is <edit-config>. which modifies the device configuration to apply the required reporting interval and temperature threshold.

c) Suppose the administre changes the reporting interval from 30 second to 10 second calculate the total Number of packets generated per minute by the 50 second before and after the change.

Ans:- Given,

* Number of sensors = 50
* Initial interval = 30 sec
* New interval = 10 sec

Before the change :-

Packets per sensor per minute = 60 ÷ 30 = 2

Total packet = 50 × 2 = 100 packets/minute

After the change :-

Packet per sensor per minute = 60 ÷ 10 = 6

Total packet = 50 × 6 = 300 packet/minte

3

Impact:- Network traffic increase from 100 to 300 packets per minute, a 200% increase. The increases bandwidth usage and gateway load but provids more frequent temperature updates.

d) NETCONF configration Management procedure.

Ans:- 1] Retrive configuration: use <get-config> to check the current device configuration

2] Modify configration: use <edit-config> to set the reporting interval and temperature threshold.

3] commit and verify: use <commit> to apply changes when supported, then use <get-config> to verify them. use SSH to secure NETCONF communication